

Skills Summary

Choosing a Ratio Representation

Use this reference while completing your worksheet or when studying.

1. The ratio table: multiply, never add

Rule: equivalent ratios come from multiplying *both* parts by the same number. $a : b$ and $ka : kb$ are the same ratio; $a + k : b + k$ is not.

Use a table when the second number is a friendly multiple of the first (double it, triple it, halve it). Write the multiplier above each arrow so you can see the jump.

Example: A paint mix is 4 parts blue to 3 parts white. How much white goes with 20 parts blue?

Step 1. The new blue amount is a whole-number multiple of the old one, so a ratio table is the quickest tool — find the multiplier, then apply it.

Step 2. $20 \div 4 = 5$, so the multiplier is 5; apply it to the white part: 3×5 .

$\Rightarrow$ 15 parts white

Try it: A mix is 7 parts sand to 4 parts cement. How much cement goes with 21 parts sand?

check: 12

2. The unit rate: the cost of exactly one

Rule: divide to get “per one”. Total cost $\div$ number of items = cost per item; distance $\div$ time = distance per unit of time.

Use a unit rate when the new amount is *not* a friendly multiple, or when two offers have different pack sizes. Once you know the per-one value, every other quantity is one multiplication.

Example: 4 smoothies cost 9.60 dollars. What does one cost?

Step 1. The question asks for the price of a single item, which is the definition of a unit rate — divide the total by the count.

Step 2. $9.60 \div 4 = 2.40$ dollars.

$\Rightarrow$ 2.40

Try it: 8 smoothies cost 21.60 dollars. What does one cost?

check: 2.70

3. The proportion: one unknown, one known ratio

Rule: set the two ratios equal and solve: $\frac{x}{\text{new}} = \frac{\text{old part}}{\text{old whole}}$, so $x = \text{new} \cdot \frac{\text{old part}}{\text{old whole}}$.

Keep the units in the same places on both sides — cups over cups, kilometres over kilometres. A flipped side is the most common error here.

Example: A recipe for 6 people uses 5 cups of stock. How much stock for 18 people?

Step 1. There is one unknown and one known ratio, which is exactly what a proportion is for.

Step 2. $\frac{x}{18} = \frac{5}{6}$, so $x = 18 \cdot \frac{5}{6} = 3 \cdot 5$.
 $\Rightarrow x = 15$ cups

Try it: A recipe for 8 people uses 7 cups of stock. How much for 24 people?

check: $17 = x$:reep

4. Comparing two rates fairly

Rule: two rates can only be compared once they are in the *same* form. Convert both to “per one” of the same thing, then compare.

Say which direction wins before you compare: for price per item the *smaller* number is better; for pages per minute the *larger* number is better. Equal ratios are a real possible answer.

Example: 3 pens for 4.50 dollars, or 5 pens for 7.00 dollars — which is the better buy?

Step 1. The pack sizes differ, so the totals say nothing on their own; convert both to price per pen and remember that here the smaller rate wins.

Step 2. $4.50 \div 3 = 1.50$ and $7.00 \div 5 = 1.40$, so the five-pack is cheaper per pen.
 $\Rightarrow 1.50 > 1.40$

Try it: 3 pens for 4.20 dollars, or 5 pens for 7.50 dollars? Compare the price per pen.

check: $1.40 < 1.50$ — the three-pack wins